Lecture 28

The index on the sphere

Objectives. *Lecture 22 associated an integer with the flux through a sphere. We now use that integer to count quantum zero modes of a Dirac operator. The spin connection changes the line bundle in which one chiral component lives. On the round sphere this turns the count into a polynomial calculation. The cited index theorem explains why the difference of the two chiral counts is topological.*

By the end of this lecture you will be able to:

- Specify the two chiral components and the sign of their index.
- Count the holomorphic sections of $O(m)$ for every $m \in \mathbb{Z}$.
- Compute both monopole zero-mode counts using the stated cited identifications.
- Distinguish the Dirac count from the spinless Landau-level count.

Unit 28A Index on the sphere

28.1 Spinors and the canonical line on the sphere

We use the unit round sphere with its outward orientation. We identify it with $\mathbb{C}\mathrm{P}^1$ using Lecture 20's complex coordinates z and $w = 1/z$. The complex orientation agrees with this outward orientation.

Definition 28.1 (Spinor field). A *spinor field* is a section of a spinor bundle, the geometric bundle on which the Dirac equation acts.

Its construction is a cited input (Lawson–Michelsohn, Chapter II and Appendix D). On an oriented surface it has two complex line components S^+ and S^-.

Definition 28.2 (Chirality). The *chirality* of a spinor is its grading with eigenvalues +1 and −1 on S^+ and S^-.

Spinor chirality is distinct from even and odd differential-form degree.

Definition 28.3 (Canonical bundle). The *canonical bundle* K_{S^2} is the line of holomorphic cotangent vectors.

Its local frames are $\mathrm{d}z$ and $\mathrm{d}w$. Since $\mathrm{d}w = -z^{-2}\mathrm{d}z$, absorbing the constant minus sign into one frame gives $K_{S^2} \cong O(-2)$ in Lecture 22's frame convention. We choose the positive spinor line to be its square root:

$$S^+ = K_{S^2}^{1/2} \cong O(-1), \qquad (K_{S^2}^{1/2})^{\otimes 2} \cong K_{S^2}.$$

This square-root choice is part of the cited spinor identification.

We can model these two lines by the spin eigenlines from Lecture 24. At a point $\mathbf{n} \in S^2$, the matrix $\mathbf{n} \cdot \boldsymbol{\sigma}$ has eigenvalues $+1$ and -1. The aligned eigenline has Chern number -1 and models S^+. The opposite eigenline has Chern number $+1$ and models S^-.

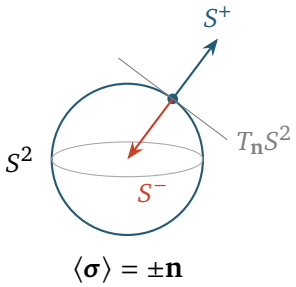
In the northern unit frame, Lecture 24 gives the aligned connection $\mathcal{A}_N = \mathrm{i}(1-\cos\theta)\mathrm{d}\varphi/2$. This is half the canonical-line connection, because the canonical line is the dual of the tangent line of Lecture 22.

The projected eigenline connections give the spin connections in this sphere model. The identification of their Dirac operator with the intrinsic operator remains the cited spinor input. A spin expectation direction alone does not establish that operator identity.

Recall that $L_n = \mathcal{O}(n)$ is the monopole line of Chern number $n \in \mathbb{Z}$. Twisting means taking the tensor product with this charged line. The positive twisted line is $S^+ \otimes L_n \cong \mathcal{O}(n-1)$. The full twisted spinor bundle has the cited description

$$S \otimes L_n \cong \left(\Lambda^{0,0} \oplus \Lambda^{0,1}\right) \otimes \mathcal{O}(n-1).$$

Here a $(0,0)$-form is a function and a $(0,1)$-form is locally a multiple of $\mathrm{d}\bar{z}$. The positive part is $\mathcal{O}(n-1)$. The negative part is $\Lambda^{0,1} \otimes \mathcal{O}(n-1)$. The figure shows the normal-spin model and the degree change under twisting.



Line	Chern number
TS^2	$+2$
K_{S^2}	-2
S^+	-1
S^-	$+1$
L_n	n
$\mathcal{O}(n-1)$	$n-1$

$$\boxed{S^+ = \mathcal{O}(-1)} \quad \otimes \quad \boxed{L_n = \mathcal{O}(n)} \quad \cong \quad \boxed{\mathcal{O}(n-1)}$$

Figure 28.1: The arrows show spin aligned with the outward normal and spin opposite to it. The degree table and the tensor product give the positive twisted line $\mathcal{O}(n-1)$.

These identifications are stated in Lawson–Michelsohn, Appendix D, equation (D.16), p. 396. Walpuski, Propositions 70.3 and 70.6, pp. 76–78, give the form decomposition and square-root construction explicitly.

28.2 The Dirac operator and zero energy

Physical input. For a charged spin-$\frac{1}{2}$ particle on the plane, minimal coupling gives $D_j = \partial_j - \mathrm{i}a(A_{\mathrm{em}})_j$, with $a = q/\hbar$. The spatial Dirac operator is $D = -\mathrm{i}(\sigma_x D_x + \sigma_y D_y)$. The massless Dirac equation is

$$\mathrm{i}\hbar\partial_t \Psi = \hbar v D\Psi, \qquad v > 0,$$

where v is the propagation speed. For a nonrelativistic particle of mass $m_{\mathrm{phys}} > 0$ with gyromagnetic factor two, the Pauli Hamiltonian is $H_{\mathrm{P}} = \hbar^2 D^2/(2m_{\mathrm{phys}})$.

The magnetic field is $B = \partial_x (A_{\mathrm{em}})_y - \partial_y (A_{\mathrm{em}})_x$. The commutator and the Pauli matrix product give

$$[D_x, D_y] = -\mathrm{i}aB, \qquad \sigma_x \sigma_y = \mathrm{i}\sigma_z,$$
$$D^2 = -D_x^2 - D_y^2 - \mathrm{i}\sigma_z [D_x, D_y] = -D_x^2 - D_y^2 - aB\sigma_z.$$

For a uniform field with $aB > 0$, choose $(A_{\mathrm{em}})_x = 0$ and $(A_{\mathrm{em}})_y = Bx$. On a coefficient $e^{\mathrm{i}ky} g(x)$, the scalar part becomes the oscillator $-g'' + (k - aBx)^2 g$. The Gaussian $g(x) = \exp[-aB(x - k/(aB))^2/2]$ has eigenvalue aB. The operators

$$b = \frac{\partial_x + aBx - k}{\sqrt{2aB}}, \qquad b^* = \frac{-\partial_x + aBx - k}{\sqrt{2aB}}$$

satisfy $[b, b^*] = 1$ and factor the scalar operator as $2aBb^*b + aB$. The Gaussian obeys $bg = 0$. Applying b^* successively therefore gives eigenvalues $(2r + 1)aB$, for integers $r \geq 0$. The spin term subtracts aB for $\sigma_z = +1$ and adds aB for $\sigma_z = -1$. The lowest orbital and spin contributions therefore cancel in the positive component. The figure records the resulting squared-Dirac energies on the plane.

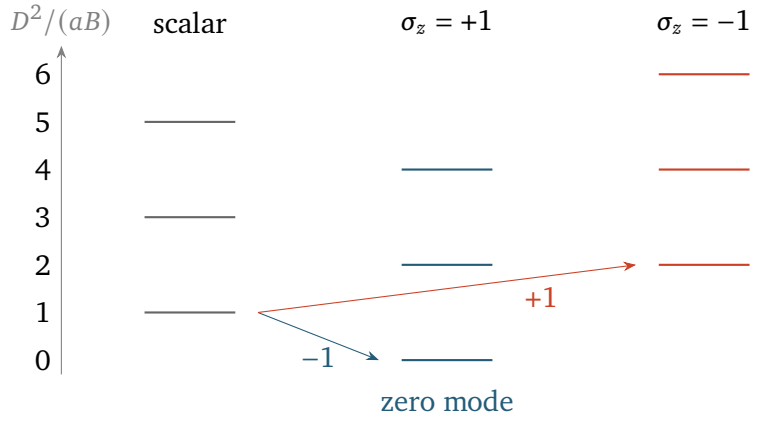


Figure 28.2: The scalar levels are shifted down or up by the spin term for $aB > 0$. The lowest positive-chirality level has zero energy; physical Pauli energies multiply these values by $\hbar^2/(2m_{\mathrm{phys}})$.

The sphere count concerns the curved Dirac operator rather than this plane oscillator spectrum. We express the curved operator using the two spinor lines.

We write $\Omega^{0,k}(\mathcal{O}(n-1))$ for the smooth $\mathcal{O}(n-1)$-valued $(0,k)$-forms.

Definition 28.4 (Dolbeault operator). The *Dolbeault operator* $\bar{\partial}_A$ is the $(0,1)$ part of the covariant derivative:

$$\bar{\partial}_A \colon \Omega^{0,0}(\mathcal{O}(n-1)) \longrightarrow \Omega^{0,1}(\mathcal{O}(n-1)).$$

In a holomorphic frame it sends a coefficient f to $(\partial_{\bar{z}} f)\mathrm{d}\bar{z}$, as in Lecture 22. Its adjoint $\bar{\partial}_A^*$ is defined by the Hermitian L^2 inner products from the bundle metric and sphere area. The first slot of these inner products is antilinear. There are no $(0,2)$-forms on a complex curve.

Proposition 28.5 (The sphere Dirac operator · *sketched*). *With the above chirality, the twisted Dirac operator is*

$$D = \sqrt{2}(\bar\partial_A + \bar\partial_A^*) = \begin{pmatrix} 0 & D^- \\ D^+ & 0 \end{pmatrix}, \qquad D^+ = \sqrt{2}\bar\partial_A, \qquad D^- = (D^+)^* = \sqrt{2}\bar\partial_A^*.$$

The block D^+ maps $\Omega^{0,0}(\mathcal{O}(n-1))$ to $\Omega^{0,1}(\mathcal{O}(n-1))$. The other block maps in the reverse direction.

Explanation. The cited spinor identification turns Clifford multiplication into exterior multiplication by an antiholomorphic covector and its adjoint contraction. Contracting the spin connection and monopole connection with it gives the displayed operator. This identification, including its $\sqrt{2}$ normalization, is Walpuski's Proposition 70.3. For a flat metric $\mathrm{d}x^2 + \mathrm{d}y^2$, the unit antiholomorphic coframe is $\mathrm{d}\bar z/\sqrt{2}$. Thus the positive block has coefficient $2\partial_{\bar z}$ and the negative block has coefficient $-2\partial_z$. Their product is $-4\partial_z\partial_{\bar z} = -\partial_x^2 - \partial_y^2$, the nonnegative scalar Laplacian of Lecture 15. To compare with the familiar flat Dirac equation, let σ_x and σ_y be the Pauli matrices. The corresponding self-adjoint spatial operator is $-\mathrm{i}(\sigma_x\partial_x + \sigma_y\partial_y)$. Multiplying its negative component by i gives the displayed chiral blocks. This verifies the local normalization of the cited operator identification. ∎

Definition 28.6 (Zero mode). A *zero mode* is a smooth spinor annihilated by D.

For a positive spinor in a holomorphic frame, the zero-mode equation is

$$D^+\psi = 0 \quad \Longleftrightarrow \quad \partial_{\bar z} f = 0.$$

This is the Cauchy–Riemann equation for the local coefficient f. Its solutions are holomorphic sections of the twisted line $\mathcal{O}(n-1)$. The transition between charts determines which local solutions extend over the sphere.

28.3 Polynomial sections and monopole zero modes

Definition 28.7 (Holomorphic sections and Dolbeault cohomology). For a holomorphic line E, the space of global holomorphic sections is $H^0(E) = \ker \bar\partial_E$. Its degree-one Dolbeault cohomology is

$$H^1(E) = \Omega^{0,1}(E)/\bar\partial_E\Omega^{0,0}(E).$$

Every $(0,1)$-form is Dolbeault-closed in complex dimension one, explaining this quotient. The cited Dolbeault Hodge theorem identifies each class with a unique harmonic representative. In this dimension that representative is annihilated by $\bar\partial_E^*$. Consequently the cited spinor and Hodge inputs identify $\ker D^+ \cong H^0(\mathcal{O}(n-1))$ and $\ker D^- \cong H^1(\mathcal{O}(n-1))$ (Walpuski, Proposition 70.3, p. 77).

We now count H^0 directly, including the global condition at infinity. Let $m \in \mathbb{Z}$ and recall the frames $e_S = e_N z^m$ for $\mathcal{O}(m)$. A section with coefficients f_N and f_S satisfies

$$f_S(w) = w^m f_N(1/w).$$

Both coefficients must be holomorphic at their own chart origin. The figure displays the monomial test for this condition.

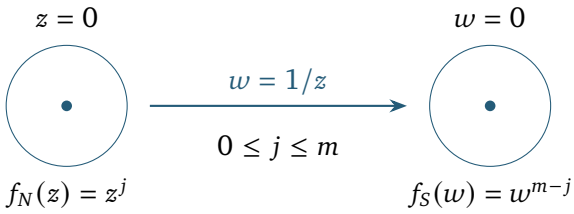


Figure 28.3: A northern monomial z^j becomes w^{m-j} in the southern frame. Holomorphicity at both poles requires $0 \le j \le m$.

We derive the needed expansion of a holomorphic function from Stokes' theorem of Lecture 12. A function is *holomorphic* here if it is smooth and satisfies $\partial_{\bar z} f = 0$. An *entire* function is holomorphic on all of $\mathbb{C}$.

Let f be holomorphic on a neighbourhood of the closed disc $|\zeta| \le R$. For a point z inside the disc, the one-form $f(\zeta)\mathrm{d}\zeta/(\zeta - z)$ is closed away from z. Indeed its exterior derivative is the $\partial_{\bar\zeta}$ derivative of its coefficient times $\mathrm{d}\bar\zeta \wedge \mathrm{d}\zeta$, which vanishes.

We apply Stokes on the disc with a small disc about z removed. The inner boundary is clockwise, so the outer counter-clockwise integral equals the counter-clockwise small-circle integral.

On the small circle $\zeta = z + \varepsilon e^{\mathrm{i}t}$, this last integral is $\mathrm{i}\int_0^{2\pi} f(z + \varepsilon e^{\mathrm{i}t})\mathrm{d}t$. Continuity and the limit $\varepsilon \to 0$ give Cauchy's formula:

$$f(z) = \frac{1}{2\pi\mathrm{i}} \oint_{|\zeta|=R} \frac{f(\zeta)}{\zeta - z}\mathrm{d}\zeta, \qquad |z| < R.$$

For $|z| < R$, the geometric series for this kernel converges uniformly on the integration circle. Termwise integration therefore gives the Taylor series and its coefficient bound:

$$\frac{1}{\zeta - z} = \sum_{j\ge 0} \frac{z^j}{\zeta^{j+1}}, \qquad f(z) = \sum_{j\ge 0} a_j z^j,$$
$$a_j = \frac{1}{2\pi\mathrm{i}} \oint_{|\zeta|=R} \frac{f(\zeta)}{\zeta^{j+1}}\mathrm{d}\zeta, \qquad |a_j| \le R^{-j} \max_{|\zeta|=R} |f(\zeta)|.$$

For an entire function, Stokes on an annulus makes each a_j independent of R. Hence this series represents the function on every disc.

Proposition 28.8 (Sections of a degree-m line · *proved*). *For $m \ge 0$, the holomorphic sections of $\mathcal{O}(m)$ have northern coefficients equal to the polynomials of degree at most m. Their dimension is $m + 1$. For $m < 0$ there is no nonzero holomorphic section.*

Proof. The northern coefficient is entire, so the Stokes–Cauchy argument gives $f_N(z) = \sum_{j\ge 0} a_j z^j$ on all of $\mathbb{C}$. The southern coefficient is bounded by some $C > 0$ on a sufficiently small closed disc about zero. By the coefficient relation, the northern coefficient satisfies

$$f_N(z) = z^m f_S(1/z), \qquad |f_N(z)| \le C|z|^m \quad \text{for sufficiently large } |z|.$$

For all sufficiently large R, the Cauchy estimate gives $|a_j| \le CR^{m-j}$. Letting $R \to \infty$ shows that $a_j = 0$ whenever $j > m$. For $m < 0$ this kills every coefficient. For $m \ge 0$ it leaves exactly $1, z, \dots, z^m$. Conversely each of these monomials gives a nonnegative southern power and hence a global section. Their distinct Taylor coefficients prove linear independence, so the dimension is $m + 1$. ∎

We can also see the negative-chirality equation directly by integration by parts. In a holomorphic frame e_N of $E = \mathcal{O}(m)$, put $h = \|e_N\|^2$ and write a negative spinor as $u = f\mathrm{d}\bar{z}\otimes e_N$. For a conformal metric $\rho(\mathrm{d}x^2 + \mathrm{d}y^2)$, the area form is $\rho\mathrm{d}x\mathrm{d}y$ and $|\mathrm{d}\bar{z}|^2 = 2/\rho$. For a test section ge_N supported in this chart, these factors cancel:

$$\langle \bar{\partial}_E(ge_N), u\rangle = 2\int \overline{\partial_{\bar{z}} g}\, fh\mathrm{d}x\mathrm{d}y = -2\int \bar{g}\,\partial_z(fh)\mathrm{d}x\mathrm{d}y.$$

Thus $\bar{\partial}_E^* u = 0$ exactly when $v = h\bar{f}$ is holomorphic. The frame and covector transitions give

$$f_S = -\bar{z}^2 z^{-m} f_N, \qquad h_S = |z|^{2m} h_N, \qquad v_S = -w^{-m-2} v_N.$$

After absorbing the constant minus sign into a frame, this is the coefficient law of $\mathcal{O}(-m-2)$. The correspondence is antilinear and bijective: its inverse is $f = h^{-1}\bar{v}$ in each chart. It explains locally why the negative count equals $\dim H^0(\mathcal{O}(-m-2))$. For example, when $n = -2$, the positive twisted degree is $m = -3$ and the dual degree is one. The choices $v = 1$ and $v = z$ give the negative coefficients $(1+|z|^2)^{-3}\mathrm{d}\bar{z}$ and $\bar{z}(1+|z|^2)^{-3}\mathrm{d}\bar{z}$ in the holomorphic frame.

The corresponding global cohomology statement is *Serre duality*, a cited theorem for compact complex curves, whose sphere case is $H^1(E) \cong H^0(K_{S^2}\otimes E^*)^*$. Here E^* is the dual line and the outer star denotes the dual vector space. See Walpuski, Remark 70.5, p. 77, or Gunning, *Some Topics in the Function Theory of Compact Riemann Surfaces*, preliminary version (2018), Theorem 1.17, p. 28. For $E = \mathcal{O}(m)$ this gives $\dim H^1(\mathcal{O}(m)) = \dim H^0(\mathcal{O}(-m-2))$. The polynomial result therefore determines both counts.

Theorem 28.9 (Monopole zero modes · *sketched*)**.** *On the round sphere with the monopole connection on $L_n = \mathcal{O}(n)$, for every $n \in \mathbb{Z}$,*

$$\dim\ker D^+ = \max(n,0), \qquad \dim\ker D^- = \max(-n,0),$$
$$\dim\ker D^+ - \dim\ker D^- = n.$$

The total number of zero modes is $|n|$. For $n > 0$ they all have positive chirality. For $n < 0$ they all have negative chirality. For $n = 0$ both kernels vanish for this sphere operator.

Proof. The positive component is $\mathcal{O}(n-1)$, so the polynomial count gives $\max(n,0)$. Serre duality identifies the negative count with $\dim H^0(\mathcal{O}(-n-1)) = \max(-n,0)$. Subtracting gives n for every integer n. Adding gives $|n|$. The global kernel identifications are the cited analytic steps in this derivation. ∎

Example 28.10 (Three zero modes in a monopole field)**.** We take the sphere monopole with $n = 3$ and the nonnegative squared operator $H = D^2$. The physical Pauli Hamiltonian is a positive multiple of this operator, so their kernels agree. We determine its zero-energy states and their chirality. For a smooth spinor Ψ, the identity $\langle \Psi, D^2\Psi\rangle = \|D\Psi\|^2$ shows that $\ker H = \ker D$. The theorem gives three positive modes and no negative mode. Here the positive line has $m = n - 1 = 2$.

For a general $m \geq 0$, use Lecture 22's unit frame for $\mathcal{O}(m)$. The local positive zero-mode equation and its monomial solutions are

$$\left(\partial_{\bar{z}} + \frac{mz}{2(1+|z|^2)}\right)\psi = 0, \qquad \psi_j(z) = (1+|z|^2)^{-m/2} z^j, \quad 0 \leq j \leq m.$$

Differentiating the weight in ψ_j gives $-mz\psi_j/[2(1+|z|^2)]$, which cancels the connection term. These are unit-frame coefficients of sections of $\mathcal{O}(n-1)$. They are not coefficients

relative to $dz^{1/2}$ or to a coordinate density. The spin connection has already entered through the degree shift from n to m. For $m = 2$ the northern coefficients are

$$\frac{1}{1+|z|^2}, \qquad \frac{z}{1+|z|^2}, \qquad \frac{z^2}{1+|z|^2}.$$

Their southern holomorphic coefficients are w^2, w and 1, respectively, so each section is holomorphic at the south pole.

We normalize these states using the round area $dA = 4dx\,dy/(1+|z|^2)^2$, with $z = x + iy$. Here dA is an area form, whereas the subscript A in $\bar\partial_A$ names a connection. The angular integral makes distinct monomials orthogonal. With $r = |z|$, the three radial norms are

$$8\pi \int_0^\infty \frac{r^{2j+1}}{(1+r^2)^4} dr = \begin{cases} 4\pi/3, & j = 0, 2, \\ 2\pi/3, & j = 1. \end{cases}$$

Substitution $t = r^2$ reduces these to the integrals of $4\pi t^j/(1+t)^4$. Next put $u = (1+t)^{-1}$. The three integrals divided by 4π become $\int_0^1 u^2 du = 1/3$, $\int_0^1 u(1-u)du = 1/6$ and $\int_0^1 (1-u)^2 du = 1/3$, giving the displayed values. Dividing each coefficient by the square root of its stated squared norm gives an orthonormal basis. All three are zero-energy states of positive chirality.

We can locate the three normalized states on the sphere. With $z = \tan(\theta/2)e^{i\varphi}$, their unnormalized coefficients are $\cos^2(\theta/2)$, $\frac{1}{2}\sin\theta e^{i\varphi}$ and $\sin^2(\theta/2)e^{2i\varphi}$. Their normalized pointwise densities are

$$\varrho_0 = \frac{3(1+\cos\theta)^2}{16\pi}, \qquad \varrho_1 = \frac{3\sin^2\theta}{8\pi}, \qquad \varrho_2 = \frac{3(1-\cos\theta)^2}{16\pi}.$$

Each density integrates to one. The first is largest at the north pole, the second at the equator and the third at the south pole. The sum is $3/(4\pi)$, independent of position. The figure uses these densities with a common scale.

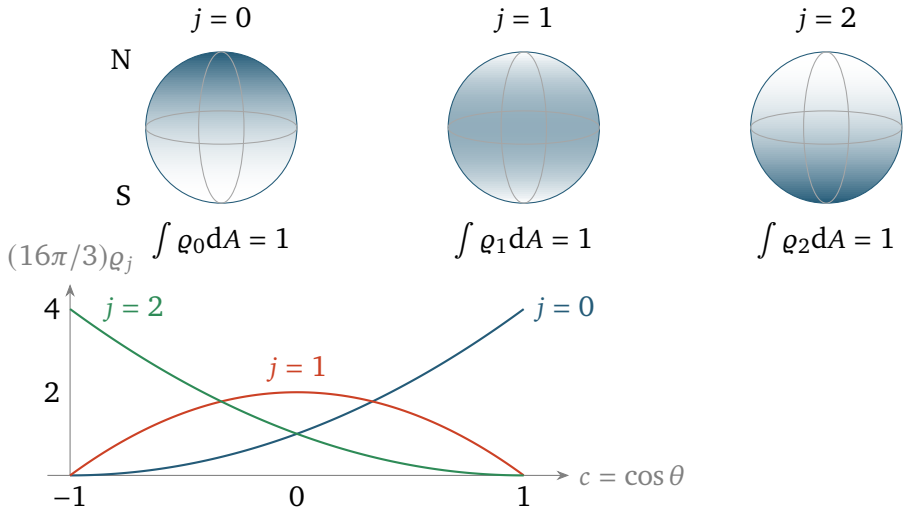


Figure 28.4: The shading and latitude curves show the three normalized $n = 3$ mode densities. The common plotted scale is $(16\pi/3)\varrho_j$, and each mode has integral one.

Rotations of the sphere lift to the monopole line through SU(2). To see their action on the positive zero modes for $n > 0$, write $m = n - 1$ and homogenize a polynomial:

$$F(Z_0, Z_1) = Z_0^m f_N(Z_1/Z_0) = \sum_{j=0}^{m} a_j Z_0^{m-j} Z_1^j.$$

Linear SU(2) substitutions in (Z_0, Z_1) preserve this degree-m space. On the monomial $Z_0^{m-j} Z_1^j$, the diagonal spin generator has weight $m/2 - j$. The operators $Z_0 \partial_{Z_1}$ and $Z_1 \partial_{Z_0}$ connect adjacent monomials with coefficients j and $m - j$. Thus these monomials form the symmetric-power spin-$m/2$ representation, with $m + 1$ states. For $n = 3$ the zero modes therefore form a spin-one triplet under these lifted rotations.

Definition 28.11 (Lowest Landau level)**.** The *lowest Landau level* is the lowest scalar magnetic-kinetic eigenspace.

Physical input. For a spinless particle on the round sphere with nonnegative monopole magnitude n, the lowest Landau level is represented by holomorphic sections of $\mathcal{O}(n)$. This sphere model is described by Greiter, §II, equations (20)–(21).

The polynomial count therefore gives $n + 1$ scalar states, compared with n spinor zero modes for $n > 0$. For $n = 3$, the scalar coefficients have degrees zero through three, giving four states. The spin connection reduces the degree by one in the Dirac problem. It cancels the +1 in the sphere's polynomial dimension formula. This comparison concerns degeneracy; the scalar lowest Landau energy is not the Dirac zero energy.[1]

28.4 The surface index theorem

The square D^2 preserves chirality. Its positive-energy states pair across the two components: if $D^- D^+ u = \epsilon u$ with $\epsilon > 0$, then $D^+ u/\sqrt{\epsilon}$ is a normalized state of $D^+ D^-$ with the same eigenvalue. Its norm is one because $\|D^+ u\|^2 = \langle u, D^- D^+ u \rangle = \epsilon$ for $\|u\| = 1$. The reverse map is $D^-/\sqrt{\epsilon}$. The figure shows why a difference can remain only at zero.

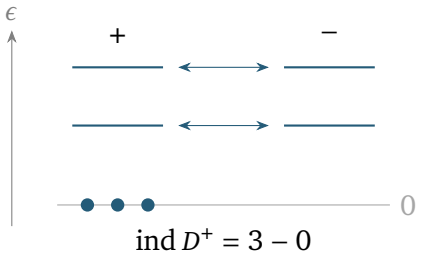


Figure 28.5: Positive squared-Dirac eigenvalues pair between chiralities. The schematic zero eigenspace has three positive modes and no negative mode, so its index is three.

The kernels are finite-dimensional by the cited elliptic theory of Lawson–Michelsohn, *Spin Geometry*, Chapter III, §5.

[1] Greiter uses negative electron charge and monopole magnitude $2s_0$. His lowest level has angular momentum s_0 and dimension $2s_0 + 1$. Here $n = 2s_0$ in the scalar comparison is a flux magnitude rather than a signed Chern convention.

Definition 28.12 (Chiral index)**.** The *index* is the signed difference

$$\operatorname{ind} D^+ = \dim\ker D^+ - \dim\ker D^-.$$

Both dimensions are over $\mathbb{C}$.

For the monopole connection, the computed kernel dimensions give $\operatorname{ind} D^+ = n$. An index of zero alone does not say that either kernel vanishes.

The polynomial count determines the left side of an index theorem for this sphere connection. The right side is the flux integer already computed in Lecture 22.

Theorem 28.13 (Atiyah–Singer on a surface · *cited*)**.** *Let M be a closed oriented Riemannian spin surface and L a Hermitian line bundle with unitary connection of curvature F_{conn}. Choose positive spinors in the square root of the canonical line for the chosen spin structure. For the corresponding chiral twisted Dirac operator,*

$$\operatorname{ind} D^+ = \int_M c_1(L) = \frac{\mathrm{i}}{2\pi}\int_M F_{\mathrm{conn}}.$$

This is the surface specialization of the twisted Dirac index theorem (Nakahara, equations (12.85)–(12.86)). The curvature here is anti-Hermitian, as in Lecture 22. For the sphere monopole, $F_{\mathrm{conn}} = -\mathrm{i}n\sin\theta\,\mathrm{d}\theta\wedge\mathrm{d}\varphi/2$. Its curvature integral gives $\mathrm{i}(-2\pi\mathrm{i}n)/(2\pi) = n$, agreeing with the kernel count. The theorem applies beyond this symmetric monopole connection. It fixes the difference of the kernels under smooth changes of the metric and connection on the same graded bundle. It does not generally fix either kernel dimension separately.

We use the self-adjoint realization of D on the closed sphere with the first-order Sobolev domain defined next. This is supplied by cited elliptic spectral theory (Lawson–Michelsohn, Theorem 5.8, pp. 196–197), with the domain identified by the elliptic estimate in Theorem 5.2.

Definition 28.14 (First-order Sobolev domain)**.** The first-order Sobolev domain consists of square-integrable sections whose first derivatives, understood by integration by parts, are also square-integrable in local charts.

Definition 28.15 (Fredholm operator)**.** An operator between the first-order Sobolev domain and L^2 is *Fredholm* if it has closed range and finite-dimensional kernel and cokernel.

The chiral block is Fredholm by the cited elliptic theory. The cokernel is the target modulo the range, and is identified with $\ker D^-$ by the adjoint. These analytic facts follow from cited elliptic theory (Lawson–Michelsohn, Chapter III, Theorem 5.2, pp. 193–194).

Along a norm-continuous path of Fredholm chiral blocks from the fixed Sobolev domain to L^2, the index is constant by the cited Fredholm stability theorem (Lawson–Michelsohn, Proposition 7.1 and Corollary 7.4, pp. 201–202). Each kernel dimension can change, with equal changes in the two counts.

The finite-dimensional model $T_t\colon \mathbb{C}^2 \to \mathbb{C}$, given by $T_t(a,b) = ta$, illustrates this: for $t \neq 0$ its kernel and adjoint kernel have dimensions 1 and 0. At $t = 0$ they have dimensions 2 and 1. Their difference stays one. This finite-dimensional family illustrates index stability. The separate counts in the symmetric sphere family were fixed by the explicit count. Perturbations that destroy the chiral block structure are outside this statement.

Self-check. For $n = 0$, determine the sphere index and both zero-mode counts. Explain what follows from the index alone.

Answer. The sphere theorem gives index zero and both kernel dimensions zero. The index alone says only that the two dimensions agree. Their vanishing uses the sphere-specific polynomial and Serre-duality count.

Self-check. For $n = -3$, determine both chiral counts and give northern negative-mode coefficients in a holomorphic frame. Also decide whether a northern coefficient z^5 is a global positive mode when $n = 5$.

Answer. The positive degree is -4, so there are no positive modes. The adjoint equation uses $O(2)$, whose coefficients are $1, z, z^2$. Since $h = (1+|z|^2)^4$, the three negative coefficients are $(1+|z|^2)^{-4}\bar{z}^j d\bar{z}$, for $j = 0, 1, 2$. For $n = 5$ the positive degree is four. The proposed z^5 becomes w^{-1} in the southern frame and is not holomorphic there.

Self-check. State in one sentence what the chiral index is and why we introduced it.

Answer. The chiral index is the difference of the complex zero-mode dimensions, introduced because positive squared-Dirac eigenvalues pair across chiralities and only the zero modes can leave an imbalance. The surface index theorem identifies that difference with the curvature integer.

Topological counts across the course

A continuous integer-valued function on a connected parameter space is constant. We used this fact to prove stability once an integer had been defined. Lifts, Stokes' theorem and comparison on overlaps gave concrete ways to compute several of these integers. The circle row concerns a smooth map $f : S^1 \to S^1$. The Euler row concerns a closed oriented surface Σ. The first Chern row concerns a line bundle L over the sphere with the northern-boundary equator convention. The band row concerns Lecture 25's two-band direction map $\widehat{\mathbf{d}}$. The instanton row uses Lecture 23's SU(2) clutching map h_{cl}. The index row concerns the closed spin surface M and twisted Dirac operator of the surface theorem. The figure records the corresponding equalities.

$\mathrm{wind}(f)$	$=$	$\deg f$	Lecture 3, 14
$\chi(\Sigma)$	$=$	$\frac{1}{2\pi}\int_\Sigma K\,dA$	Lecture 6, 18
$\int_{S^2} c_1(L)$	$=$	$\mathrm{wind}(g_{NS})$	Lecture 22
C	$=$	$\deg \widehat{\mathbf{d}}$	Lecture 25
k	$=$	$-\deg h_{\mathrm{cl}}$	Lecture 23
$\mathrm{ind}\, D^+$	$=$	$\int_M c_1(L)$	Lecture 28

Figure 28.6: Each equality connects two descriptions of the same count, under the hypotheses printed in the cited lectures. The instanton row uses the clutching map and its stated equator orientation.

Winding in Lecture 3 records the endpoint of a lifted circle loop. Degree in Lecture 14 counts signed preimages between closed connected oriented manifolds of the same dimension. For sphere maps, Lecture 26 also identifies degree with the corresponding sphere homotopy class.

The Euler characteristic joins homology, indices of zeros and total curvature in Lectures 6, 14 and 18. The line-bundle integer joins transition winding and curvature in Lecture 22. Lecture 25 applies the same construction to a band eigenline. In Lecture 23 the instanton integer is the degree of the asymptotic $S^3 \to \mathrm{SU}(2)$ map, with the opposite sign for the clutching map.

Defect data in Lecture 26 are homotopy classes, sometimes with a fundamental-group action; they need not be integers. The kink and vortex of Lecture 27 have integer charges under their stated boundary conditions.

The sphere Dirac calculation adds a spectral count to this list. Its index fixes the difference of zero-mode dimensions, while the separate dimensions come from the explicit polynomial and duality calculation. Berry holonomy remains a phase rather than an integer.

Unit 28B Chiral anomaly

28.5 The two-dimensional chiral anomaly *(optional)*

The sphere count, the surface index theorem and the closing synthesis do not depend on this section.

For this application we reinterpret two dimensions as Euclidean spacetime. The spatial-first coordinates are (x, τ) with orientation $\mathrm{d}x \wedge \mathrm{d}\tau$ and $z = x + \mathrm{i}\tau$. The Wick rotation is $t = -\mathrm{i}\tau$, so the local holomorphic argument becomes $z = x - t$. A fixed value of $x - t$ moves toward increasing x. Positive Euclidean chirality therefore continues to right movers. This local convention matches the Euclidean grading to the Lorentzian one.

Definition 28.16 (Chiral anomaly)**.** An *anomaly* is a failure after quantization of a classical conservation law. For the massless Dirac field with vector gauge coupling, the chiral anomaly concerns the separate right- and left-moving fermion-number currents.

Remark 28.17 (Axial charge and index · *cited*)**.** Let j_R and j_L be those currents, and let Q_R and Q_L be their integrated net numbers of particles minus antiparticles. The vector and axial charges are

$$Q = Q_R + Q_L, \qquad Q_5 = Q_R - Q_L, \qquad j_5 = j_R - j_L.$$

Physical input. Classically the massless model conserves the right- and left-moving number currents separately. Quantization preserving the vector current can violate axial-current conservation in a gauge background. For one Dirac fermion, with the chirality convention above and with no axial charge escaping through a spatial boundary, the cited integrated law is

$$\Delta Q_5 = 2 \operatorname{ind} D^+, \qquad \Delta Q = 0.$$

This is the chiral anomaly input from Bilal, §3.5–3.7, with the even-dimensional extension

in equations (3.44)–(3.47).[2]

If the Euclidean index is N, the sum and difference equations give $\Delta Q_R = N$ and $\Delta Q_L = -N$. An index +1 background therefore changes the imbalance by +2, while preserving the vector charge. The Euclidean spacetime background in this statement is distinct from the spatial sphere used to count monopole states.

We illustrate the sign with a ring of circumference $\ell > 0$.

Physical input. Take noninteracting massless right and left movers with speed $v > 0$ on the ring, coupled to a spatially uniform A_x in temporal gauge. The chiral Hamiltonians are $v(-i\hbar\partial_x - qA_x)$ and its negative. The reference vacuum fills the negative-energy states. During the flux change we transport occupations of the uncoupled momentum modes. Net particles and holes are counted relative to the final negative-energy vacuum.

We impose antiperiodic boundary conditions to avoid zero-energy endpoints. The allowed momenta and dimensionless flux are

$$k_j = \frac{2\pi}{\ell}\left(j + \frac{1}{2}\right), \quad j \in \mathbb{Z}, \qquad \alpha = \frac{a\ell A_x}{2\pi}, \quad a = q/\hbar.$$

In units of $2\pi\hbar v/\ell$, the two spectra are

$$\varepsilon_{R,j} = j + \frac{1}{2} - \alpha, \qquad \varepsilon_{L,j} = -\left(j + \frac{1}{2} - \alpha\right).$$

We increase $a\ell A_x$ by $+2\pi$, so α increases from zero to one. The endpoints are gauge equivalent through the single-valued phase $e^{2\pi i x/\ell}$. The figure follows the levels without changing their occupations.

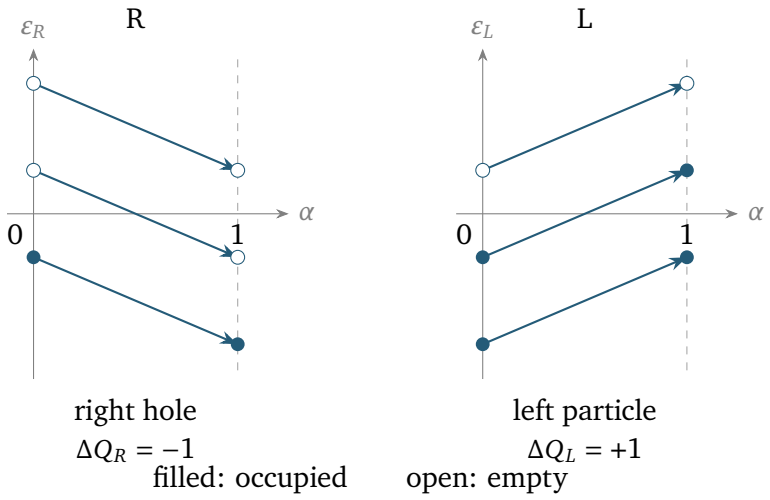


Figure 28.7: Increasing $a\ell A_x$ by 2π moves one empty right level down and one filled left level up through zero. Relative to the final negative-energy sea, they give one right hole and one left particle.

[2] Bilal's equation (3.30) writes the current divergence as minus his anomaly function. His equation (3.41) gives the integral of that function as minus twice the index in four dimensions. The dimension-independent regulated trace in equations (3.39)–(3.40) pairs nonzero modes and counts zero modes with a sign. The fermion and antifermion measures give the factor two. We use the two-dimensional consequence with positive Euclidean chirality continued to right movers.

Initially the occupied right levels have $j \le -1$, whereas the final right sea has $j \le 0$. The transported state is missing its $j = 0$ particle, so $\Delta Q_R = -1$. Initially the occupied left levels have $j \ge 0$, whereas the final left sea has $j \ge 1$. The transported state has one extra $j = 0$ particle, so $\Delta Q_L = +1$. All other occupation differences vanish. This subtraction gives finite changes without subtracting two infinite totals:

$$\Delta Q = 0, \qquad \Delta Q_5 = -2.$$

For the corresponding real Euclidean interpolation $A_x(\tau)$, the orientation is $\mathrm{d}x \wedge \mathrm{d}\tau$. Its curvature is $F_{\mathrm{em}} = -A_x'(\tau)\mathrm{d}x \wedge \mathrm{d}\tau$. Gluing the gauge-equivalent endpoints gives a spacetime torus. The surface index theorem gives

$$\operatorname{ind} D^+ = \frac{a}{2\pi}\int F_{\mathrm{em}} = -\frac{a\ell}{2\pi}\Delta A_x = -1.$$

Thus this flux direction gives $\Delta Q_5 = 2 \operatorname{ind} D^+ = -2$. Reversing the insertion gives index $+1$ and axial change $+2$.

Summary. The positive spinor line on the outward sphere is $O(-1)$. Twisting by the monopole line $O(n)$ gives $O(n-1)$. The Stokes–Cauchy polynomial proof and cited spinor, Hodge and Serre identifications give chiral counts $\max(n, 0)$ and $\max(-n, 0)$. Their difference is n, agreeing with the cited surface index theorem and the flux Chern number. For positive n, the spinor count is n and the spinless lowest Landau-level count is $n + 1$. With the stated spacetime, chirality and boundary conventions, the cited axial anomaly gives $\Delta Q_5 = 2 \operatorname{ind} D^+$. The ring insertion of $a\ell A_x$ by $+2\pi$ has index -1 and axial change -2.

Reading.

- H. B. Lawson and M.-L. Michelsohn, *Spin Geometry* (Princeton University Press, 1989), Chapter III, §5, pp. 192–197, and §7, pp. 201–202; Chapter IV, §11; Appendix D, p. 396.
- Thomas Walpuski, *Differential Geometry IV* (2022), Propositions 70.3 and 70.6 and Remark 70.5, pp. 76–78.
- R. C. Gunning, *Some Topics in the Function Theory of Compact Riemann Surfaces*, preliminary version (2018), Theorem 1.17, p. 28.
- Mikio Nakahara, *Geometry, Topology and Physics*, 2nd ed. (2003), §12.6.
- Adel Bilal, *Lectures on Anomalies* (2008), §3.5–3.7, pp. 13–17.
- Martin Greiter, *Landau Level Quantization on the Sphere*, *Phys. Rev. B* 83 (2011), 115129, §II.